Fake News Detection

Using the LIAR Dataset · ML Classification Pipeline

Models: Logistic Regression · Gradient Boosting · DistilBERT

Dataset: LIAR · ~12,800 political statements · Binary Classification (FAKE / REAL)

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Problem Statement & Dataset Overview

Problem Statement

Misinformation spreads rapidly online, making automatic fake news detection critical for media integrity.

Goal: Classify political statements as FAKE or REAL using NLP and machine learning techniques.

Input: Raw political statements with speaker metadata.

Output: Binary label — FAKE (0) or REAL (1).

LIAR Dataset

~12,800 Total Statements	**10,240** Training Samples
1,284 Validation Samples	**1,267** Test Samples

Methodology & Pipeline

1	2	3	4	5	6	7
Data Collection	**Preprocessing**	**EDA**	**Feature Eng.**	**Modelling**	**Evaluation**	**Deployment**
LIAR dataset download & loading	Clean, tokenise, lemmatize text	Label & length distributions, party	TF-IDF + linguistic + credit features	LR · GB · DistilBERT	Accuracy, F1, ROC-AUC, SHAP	Streamlit web app with UI

Label Mapping: FAKE (0) | REAL (1)

Feature Engineering

1. TF-IDF Features

- Unigram + Bigram (n-gram range 1–2)
- Max 50,000 features, sublinear TF scaling
- Min document frequency = 2
- Produces sparse matrix for classical models

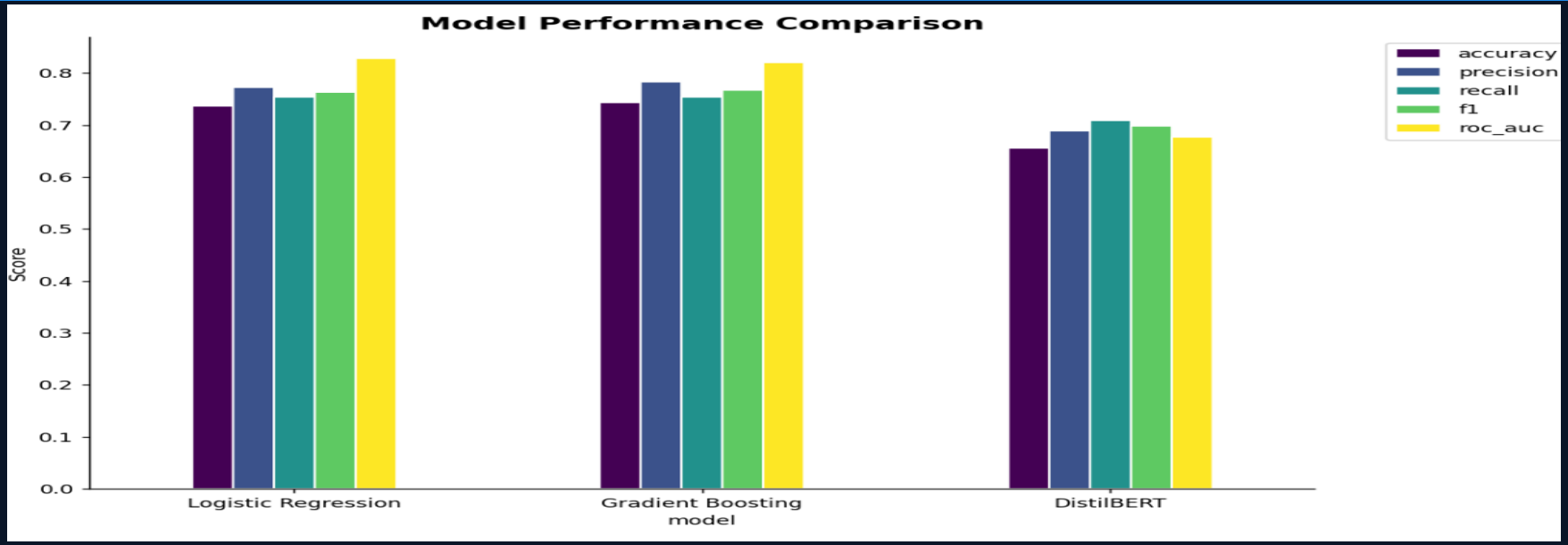
2. Linguistic Features

- Exclamation/question marks, uppercase words
- Lexical diversity, avg word/sentence length
- Hedging words, superlatives, number counts
- Flesch-Kincaid readability score

3. Credit-History Features

- Speaker's historical counts: pants-fire,
- false, barely-true, half-true, mostly-true
- Deception ratio & truth ratio (normalised)
- Total credit history count per speaker

Key Results & Model Comparison



Best Model: Gradient Boosting

Accuracy: 74.35% | F1: 76.80% | ROC-AUC: 82.08%

"DistilBERT's performance reflects the nature of the dataset rather than a limitation of transformer models a metadata-rich corpus naturally favors structured feature-based approaches."

Model Performance- Detailed Metrics

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	73.64%	77.30%	75.35%	76.31%	**82.84%**
Gradient Boosting	**74.35%**	**78.31%**	75.35%	**76.80%**	82.08%
DistilBERT	65.59%	68.94%	70.87%	69.89%	67.70%

LR

Best ROC-AUC (82.84%) excellent probability calibration for ranking tasks.

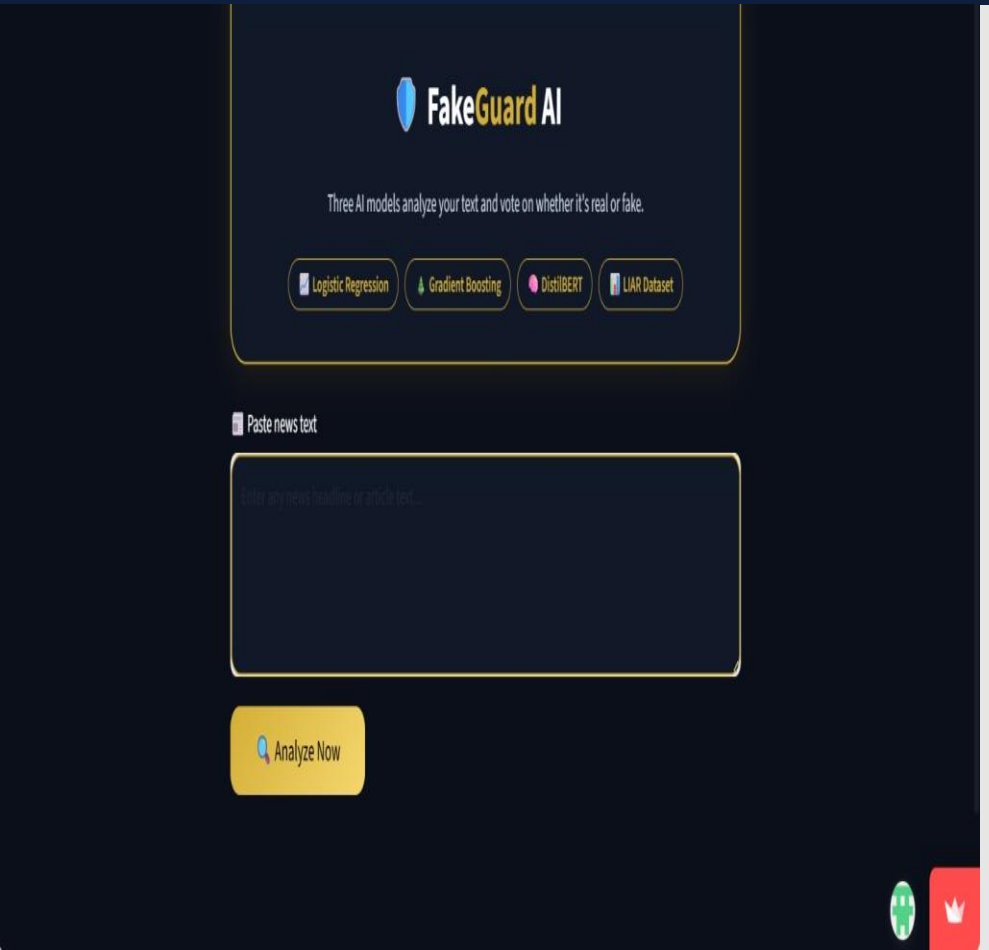
GB

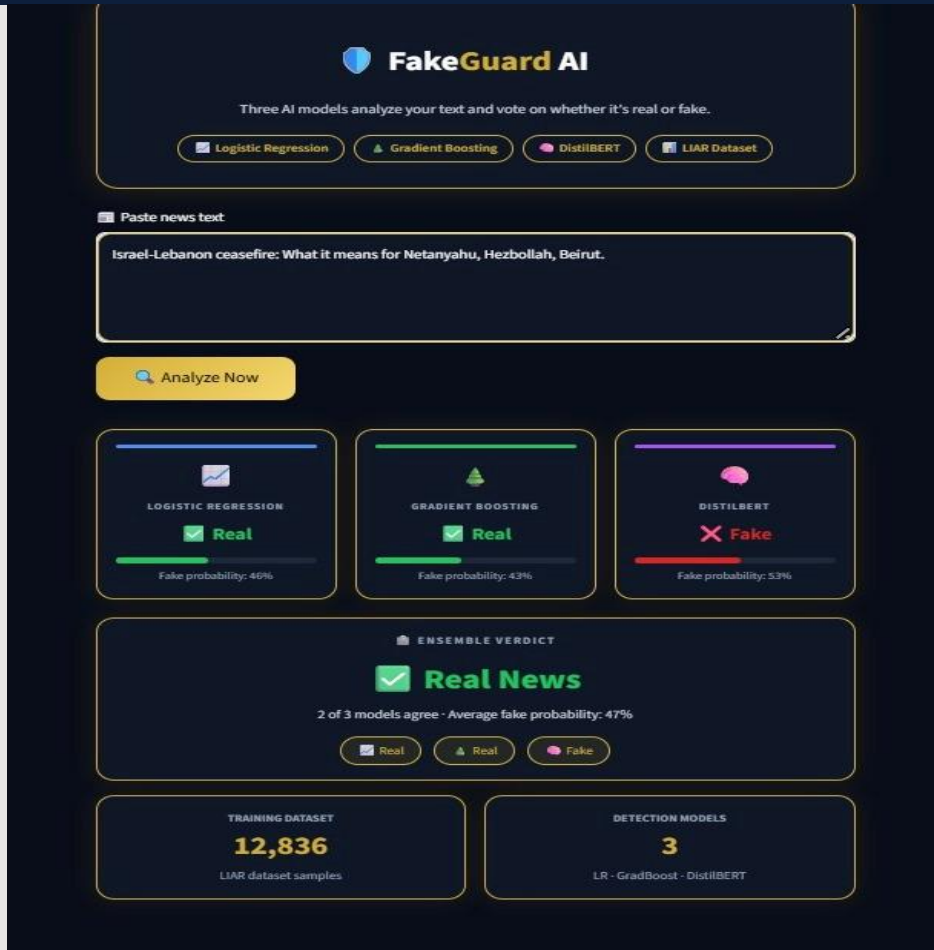
Best overall Accuracy & F1. LightGBM handles sparse TF-IDF efficiently.

BERT

Underperforms classical models limited training epochs & no metadata features.

Deployed Application- Streamlit





Conclusions, Limitations & Future Work

Conclusions

- Gradient Boosting achieved best accuracy (74.35%) and F1 (76.80%)
- Classical ML (LR + GB) with engineered features outperformed DistilBERT
- Speaker credit-history is a highly informative signal for deception
- TF-IDF + linguistic + metadata features form a strong feature set
- SHAP / LIME provide strong model interpretability

Limitations

- LIAR dataset is politics-specific may not generalise to other domains
- Binary mapping of 6 labels loses nuance (barely-true -> FAKE)
- DistilBERT trained for only 3 epochs due to compute limits
- No cross-lingual capability English statements only
- Static model not updated with new misinformation patterns

Future Work

- Fine-tune larger models: RoBERTa, DeBERTa, or GPT variants
- Incorporate knowledge graphs and fact-checking APIs
- Multi-class prediction (6 labels) with ordinal regression
- Ensemble LR + GB + BERT via soft-voting or stacking
- Real-time browser extension for live news verification